

SQL Cheatsheet for Beginners

(2025 Edition)

1.Setup + SQL Data Types

Setup: Create / Drop Database or Table

```
CREATE DATABASE demo_db;
DROP DATABASE demo_db;

CREATE TABLE customers (
  customer_id INT,
  customer_name VARCHAR(50),
  city VARCHAR(50)
);
DROP TABLE customers;

INSERT INTO customers VALUES (1, 'Alice', 'London');
```

Real-Life Data → SQL Data Types

Real-Life Example	SQL Data Type	Example Value
ID,quantity	INT	101,25
Price,salary	DECIMAL(10,2)	25000.75
Name,city	VARCHAR(50)	'London'
Date fields	DATE	'2024-06-01'
Yes/No flags	BOOLEAN	TRUE,FALSE

2.SELECT + Filtering Basics

SELECT Queries:

```
SELECT customerName, country  
FROM customers;
```

```
SELECT customerName  
FROM customers  
WHERE country = 'USA';
```

Filtering — Integer, String, Date:

-- Integer

```
WHERE creditLimit > 50000  
WHERE customerNumber BETWEEN 200 AND 300
```

-- String

```
WHERE city IN ('Paris', 'Tokyo')  
WHERE customerName NOT IN ('Land of Toys')
```

-- Date

```
WHERE orderDate = '2024-06-01'  
WHERE orderDate BETWEEN '2024-06-01' AND '2024-06-30'
```

Advanced Filters (NULL + Logic)

NULL / NOT NULL:

WHERE phone IS NULL

WHERE phone IS NOT NULL

Logical Operators:

WHERE country = 'USA' AND creditLimit > 50000

WHERE city = 'London' OR country = 'UK'

WHERE NOT (country = 'Germany')

3. Pattern Matching (LIKE)

WHERE customerName LIKE 'A%'	-- Starts with A
WHERE customerName LIKE '%ia'	-- Ends with 'ia'
WHERE customerName LIKE '%mot%'	-- Contains 'mot'
WHERE customerName LIKE '_A%'	-- 2nd char is A
WHERE customerName LIKE '__DIA'	-- 5-letter ending in 'DIA'
WHERE country LIKE 'A%'	-- Country starts with A

💡 % = any characters, _ = one character

4.Sorting + Limiting

ORDER BY:

```
ORDER BY customerName ASCORDER BY creditLimit DESC;
```

LIMIT:

```
LIMIT 5 ;
```

5.JOINs (2-table + 3-table)

2-Table JOIN

```
SELECT c.customerName, o.orderNumber
FROM customers c
JOIN orders o
    ON c.customerNumber = o.customerNumber;
```

3-Table JOIN

```
SELECT c.customerName, o.orderNumber, p.productName
FROM customers c
JOIN orders o ON c.customerNumber = o.customerNumber
JOIN orderdetails od ON o.orderNumber = od.orderNumber
JOIN products p ON od.productCode = p.productCode;
```

6.Query Mindset + Engine Flow

Query-Writing Mindset:

1. SELECT – What to retrieve?
2. FROM – Which table(s)?
3. WHERE – Any filter conditions?
4. ORDER BY – Want sorted output?
5. LIMIT – Want top N rows?
6. JOIN – Any related tables?

You've covered the foundations of SQL: SELECT, filtering, sorting, joins, and logic.

Now it's your turn:

- Play with real queries
- Explore **classicmodels** freely
- Bookmark this cheatsheet
- Share with friends — it helps you revise

SQL isn't about memory. It's about mindset.

*Created by **Chandra Venkat** · Based on classicmodels · 2025 Edition*